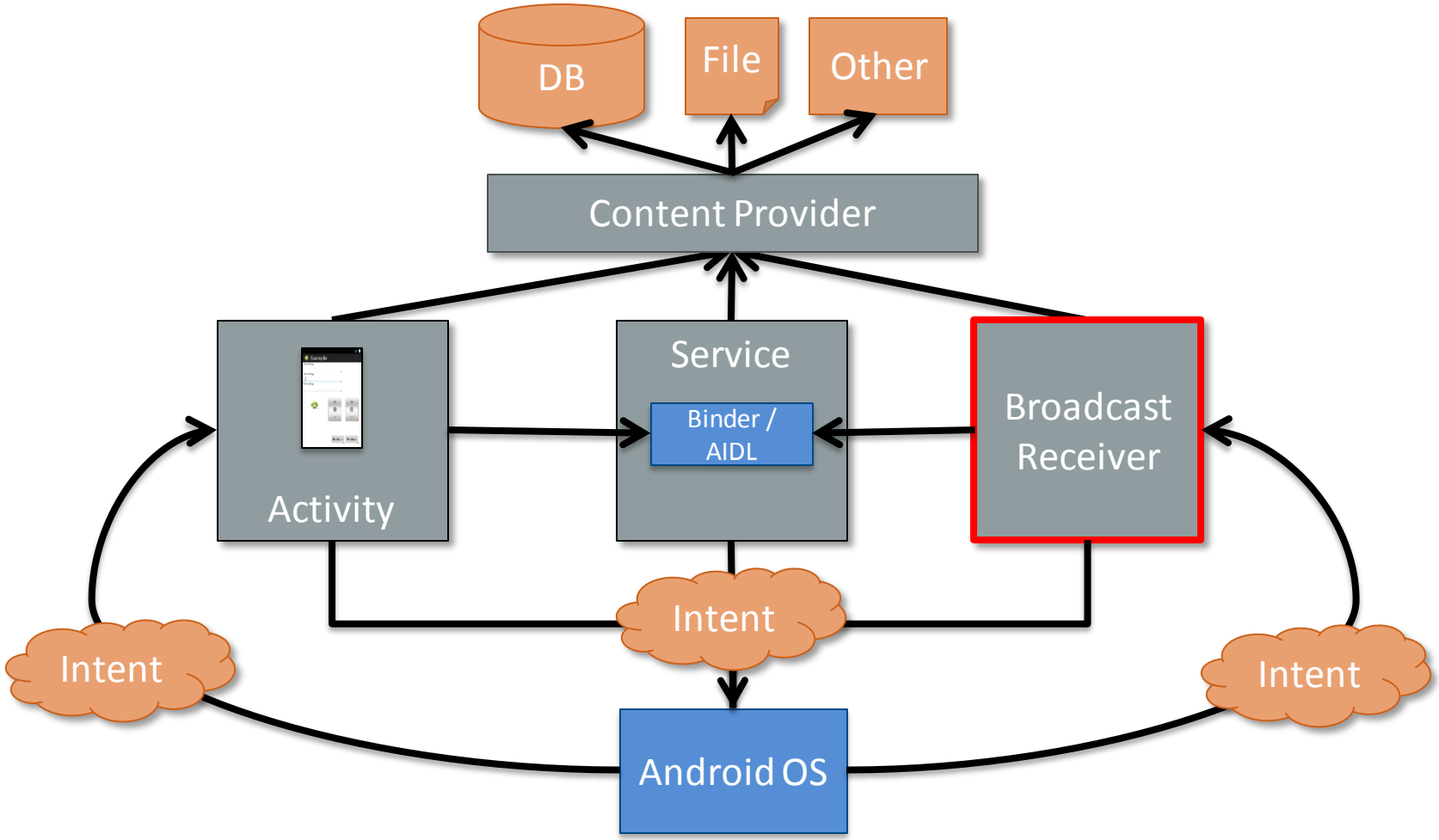
605.486 – Android Development

Module 19
Widgets and Broadcast Receivers

Application Components

and Typical Communication



Broadcast Receiver

- Intercepts broadcast intents
- *All matching receivers receive broadcast intents*
  - *(Unless canceled)*

- System intents or other broadcast intents
- Can perform *very* short processing
  - Otherwise send to service for processing

- In general, do not start activities or show dialogs/toasts
  - This example does to explicitly show the broadcast received

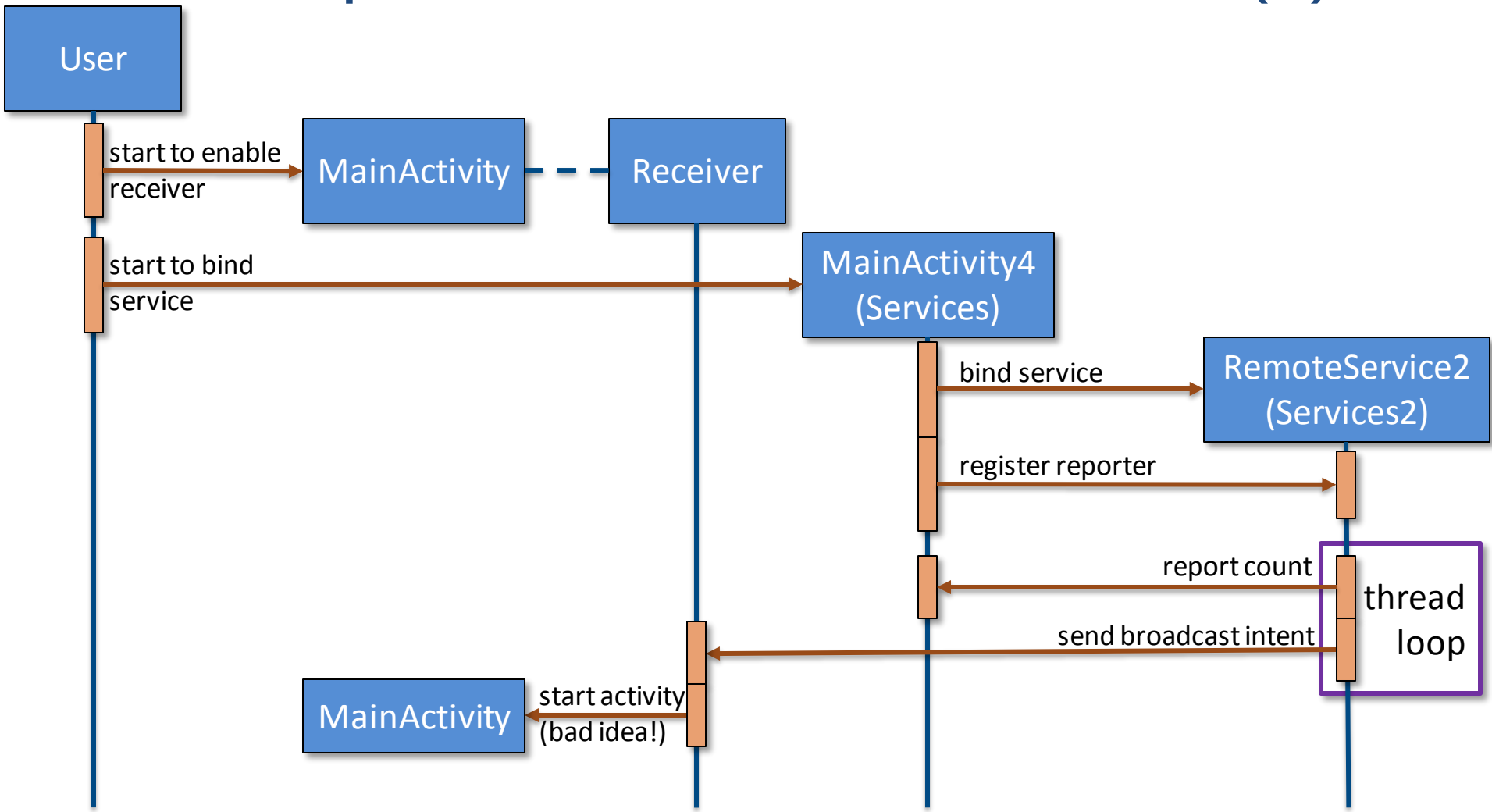
- Note: As of ICS – receivers are *not* invoked until app has been invoked at least once!

Example 1 - Broadcast Receiver (1)

- Counting service sends broadcast intents
- Receiver accepts them
- When receiver sees count hit 10
  - Start activity (bad idea… just for example)

- Note: requires user interaction to enable receiver!
  - Activity start, widget setup

Example 1 - Broadcast Receiver (2)



Example 2 - Widgets

- Limited UI on home screens
- Widget Provider – manages widget UI; is Broadcast Receiver
- Widget Layout (xml)
- Widget Info (xml)

Example 3 – Getting Fancier (1)

- Widgets can ***only*** contain
  - FrameLayout, LinearLayout, RelativeLayout, GridLayout
  - AnalogClock, Button, Chronometer, ImageButton, ImageView, ProgressBar, TextView, ViewFlipper, ListView, GridView, StackView, AdapterViewFlipper

Example 3 – Getting Fancier (2)

- Add colored blocks to our widget
  - GridLayout to manage blocks
  - Use TextView
  - Set its background color based on count
  - Remove every 10 counts
- RemoteViews can add/remove views, call setter methods

"INFO" Advice Credit

- Advice from Justin at nEx.Software
  - http://nexsoftware.net/wp/2009/07/29/tutorial-creating-a-custom-analogclock-widget/
  - Set your Activity Intent Filter to use category INFO instead of LAUNCHER
  - Hides the activity from the app drawer
  - Displays it when user presses Open in play store